

In-Class Exercise: Solutions

Hydraulic Bottle Jack, Mechanical and Hydraulic Systems

Worked from the dimensions marked on the exercise figure. Units are inches, pounds, psi and horsepower, with 1 hp = 6600 in-lb/s. All values are ideal: no friction, no leakage.

Given

W = 4000 lb, D_r = 2.00 in, d_p = 0.50 in, s_p = 0.70 in, L = 7.00 in, a = 1.00 in, b = 12.00 in, theta = 44.4 deg, D_i = 2.10 in, D_o = 3.50 in, H = 6.00 in, n = 1 cycle/s

Calculate

1. Areas

$$A_p = \pi d_p^2 / 4 = \pi (0.50)^2 / 4 = 0.1963 \text{ in}^2$$

$$A_r = \pi D_r^2 / 4 = \pi (2.00)^2 / 4 = 3.1416 \text{ in}^2$$

2. Volume displaced per stroke

$$V_p = A_p s_p = 0.1963 \times 0.70 = 0.1374 \text{ in}^3$$

3. Lift per stroke

$$h = V_p / A_r = 0.1374 / 3.1416 = 0.04375 \text{ in}$$

Conservation of volume: the oil is incompressible, so the volume leaving the pump appears under the ram. Equivalently $h = s_p (A_p / A_r) = 0.70 / 16 = 0.04375 \text{ in}$.

4. Pressure at the rated load

$$P = W / A_r = 4000 / 3.1416 = 1273.2 \text{ psi}$$

5. Force on the pump piston

$$F_p = P A_p = 1273.2 \times 0.1963 = 250.0 \text{ lb}$$

Pascal's principle: with the delivery ball open the pump chamber and the lifting chamber are one body of fluid at one pressure, so the same P acts on both pistons.

6. Torque and hand force

$$T = F_p a = 250.0 \times 1.00 = 250.0 \text{ lb-in}$$

$$F_{\text{hand}} = T / b = 250.0 / 12.00 = 20.83 \text{ lb}$$

7. Strokes, time, and speed

$$N = L / h = 7.00 / 0.04375 = 160.0 \text{ strokes}$$

$$t = N / n = 160.0 / 1 = 160.0 \text{ s}$$

$$v = L / t = 7.00 / 160.0 = 0.04375 \text{ in/s}$$

8. Power

$$Q = V_p n = 0.1374 \text{ in}^3/\text{s}$$

$$\text{Power}_{\text{hyd}} = P Q = 1273.2 \times 0.1374 = 175.00 \text{ in-lb/s} = 0.02652 \text{ hp}$$

$$\text{Power}_{\text{out}} = W v = 4000 \times 0.04375 = 175.00 \text{ in-lb/s} = 0.02652 \text{ hp}$$

$$d = 2 \pi b \theta / 360 = 2 \pi (12.00)(44.4) / 360 = 9.299 \text{ in}$$

$$\text{Power}_{\text{in}} = F_{\text{hand}} d n = 20.83 \times 9.299 = 193.73 \text{ in-lb/s} = 0.02935 \text{ hp}$$

9. Reservoir

$$A_{\text{res}} = \pi (D_o^2 - D_i^2) / 4 = \pi (3.50^2 - 2.10^2) / 4 = 6.1575 \text{ in}^2$$

$$V_{\text{lift}} = A_r L = 3.1416 \times 7.00 = 21.991 \text{ in}^3$$

$$\text{drop} = V_{\text{lift}} / A_{\text{res}} = 21.991 / 6.1575 = 3.571 \text{ in}$$

capacity = A_res H = 6.1575 x 6.00 = 36.95 in³

margin = capacity / V_lift = 36.95 / 21.991 = 1.68. The reservoir holds one full extension with 68 percent to spare, and the drop of 3.57 in fits inside the 6.00 in column.

10. Force multiplication

W / F_hand = 4000 / 20.83 = 192.0

(A_r / A_p)(b / a) = (3.1416 / 0.1963) (12.00 / 1.00) = 16.0 x 12.0 = 192.0

The two agree. Sixteen comes from the hydraulics and twelve from the lever, and they simply multiply.

Explore

Explore 1. Lifting cylinder bore 4.00 in

A_r = pi (4.00)² / 4 = 12.566 in², four times larger.

h = V_p / A_r = 0.1374 / 12.566 = 0.01094 in, one quarter of before.

N = L / h = 7.00 / 0.01094 = 640 strokes, four times as many.

P = W / A_r = 318.3 psi, F_p = 62.50 lb, F_hand = 5.21 lb, one quarter of before.

One principle covers all three: the area ratio. The same pressure over four times the area carries the load with a quarter of the force, and the same volume over four times the area rises a quarter as far. Force is multiplied by exactly what distance is divided by.

Explore 2. Pump bore 0.25 in

A_p = pi (0.25)² / 4 = 0.04909 in², one quarter of before. P is unchanged at 1273.2 psi because pressure is set by the load.

F_p = P A_p = 62.50 lb, F_hand = F_p a / b = 5.21 lb, one quarter of before.

The price: h = A_p s_p / A_r = 0.01094 in and N = 640 strokes. Halving the pump bore quarters the effort and quadruples the strokes.

Explore 3. The three powers

Power_hyd = 0.02652 hp and Power_out = 0.02652 hp are equal, as they must be: P Q = (W / A_r)(A_r v) = W v.

Power_in from the arc is 0.02935 hp, larger by 1.107.

That factor is theta / sin(theta) = 0.7749 / 0.6997 = 1.108: the hand travels an arc of b theta while the lever only needs the straight-line displacement s_p (b / a) = 8.40 in to move the rod its stroke. The ideal input power, F_hand times that straight-line travel, is 0.02652 hp, equal to the other two. The excess is geometry in the arc approximation, not a loss. Note a sin(theta) = 1.00 x 0.6997 = 0.700 in, which matches s_p = 0.70, so the three linkage measurements are consistent.